VOUCH-OPT: Who to Ask, Who to Match, When to Wait

A sequential reciprocal matching policy and its Round 1 research approach

Vouchsafe Sequential Matching Hackathon · The One Introduction Problem · Participant specification 1.0.0
Prepared 9 October 2026 (Asia/Kolkata)

Status: proposed method and planned evaluation. No experiment results, benchmark figures or performance claims are reported. All profiles and outcomes in the simulator are synthetic, and simulator performance is not evidence of real relationship prediction.

1. Summary and research thesis

VOUCH-OPT is a sequential, reciprocal matching policy. On each simulated day it chooses (i) which affordable missing information to request and, after receiving the updated observations, (ii) which fully feasible, non-overlapping introductions to propose. It may return an empty batch. Its objective is to improve Mutual Second-Meeting Intention (MSMI), not merely initial acceptance or superficial similarity [1–5].

Thesis. Make hard reciprocal feasibility a non-negotiable gate; spend clarification budget only where an answer can change the pool-wide decision; and choose each day's introductions as a global matching problem rather than a sequence of independent pair rankings. The note tests whether this combination improves the scored outcome (MSMI) without sacrificing action validity.

2. Problem interpretation and hypotheses

This is a sequential allocation problem, not a conventional recommendation list. On each day the policy first selects affordable information requests, then receives refreshed observations, and then proposes a disjoint batch of introductions or waits. Arrivals, availability, outstanding introductions, past pairs and delayed feedback alter later choices. An introduction is appropriate only if both people are currently eligible and able to act; a high soft-affinity or outcome score can never repair a hard-constraint failure [1–3].

The primary target is one qualifying introduction: a date occurs within 30 simulation days after assignment, and both members report "yes" to meeting again within three simulation days of that date. This sits downstream of assignment, two directional responses, mutual acceptance, a date and two timely second-meeting reports, and is not interchangeable with any earlier event [1].

Daily decision loop.

1 **Observe.** Current members, availability, past pairs and pending feedback.
2 **Ask.** Select high-value disclosures under the daily budget.
3 **Allocate.** Refresh state; match on fully feasible edges, or abstain.

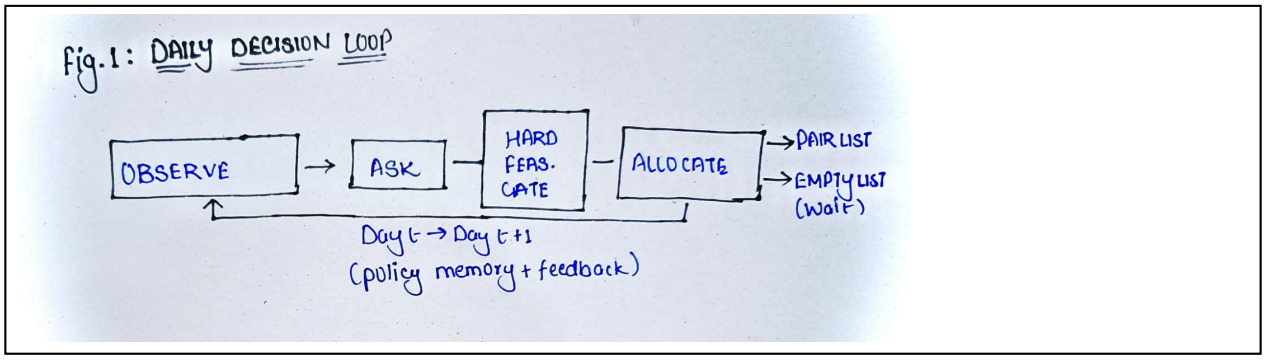


Figure 1. The daily decision loop (observe, ask, allocate), with the state that carries between days. *Space reserved for a hand-drawn diagram. Suggested content: the three phases as boxes, the hard feasibility gate before allocation, and the arrow from day t to day t+1 carrying policy memory.*

Hypotheses to test (not findings).

- **H1, pool-level allocation.** Given the same feasible edges and edge values, maximum-weight general-graph matching yields a higher mean MSMI per 100 arrived members than greedy highest-edge-first allocation, because it can avoid consuming a person needed for a better set of non-overlapping pairs. The specification's A–D / B–C example illustrates this structure; its values are not empirical estimates [1].
- **H2, targeted clarification.** Selecting clarifications by estimated decision value per budget unit outperforms no clarification on MSMI when missing hard information blocks otherwise promising feasible options. The gain may vanish when answers are declined, sparse geography leaves few candidates, or an answer would not change the selected matching.
- **H3, sequential abstention.** A tuned, explicit wait/reservation threshold may avoid low-value early matches that block better pool allocations, while still allowing a match when remaining opportunities or time make waiting less attractive. This concerns the public simulator only; no real-world relationship claim follows.

Guardrail. Every assessed episode must be valid. The approach must not trade constraint correctness for score, and every reported result includes validity, coverage, cost and the outcome funnel, not only MSMI [1].

3. Hard reciprocal feasibility gate

Construct the candidate set from distinct, currently available adults in the same pool. Apply the supplied reference eligibility semantics in both directions and do not infer unstated preferences [1, 2]. The kit's rules are used as given, not recreated or weakened.

Reciprocal eligibility	Operational eligibility
Stated age ranges and `who_to_meet`; relationship-structure compatibility; smoking and children-related rules; each person accepts the other's current fictional zone; at least one shared schedule slot.	Distinct identifiers, same pool, both present and currently available; no outstanding introduction for either person; no repeated pair in the episode; each person appears at most once in today's batch.

Three-state classification. A pair is *feasible* only when all required checks pass; *infeasible* when any known hard rule fails; and *needs clarification* when no known failure exists but required information is missing. A known hard failure takes precedence over missing fields. Unknown or declined answers never count as consent [1, 2].

Age and gender are not used to invent a preference. A zone is a supplied allocation constraint, not a real distance, and relocation interest does not override current zone acceptance. Missing or declined values are not negative preferences [1, 2].

4. Why the decisions are coupled

One chosen introduction consumes two members' current opportunities, so ranking pairs independently can waste a better combination. Information requests must likewise be assessed by their effect on the best achievable batch, not by how many profile fields they fill. Global matching solves the current day's allocation; abstention manages the unresolved trade-off with future opportunities [6–9].

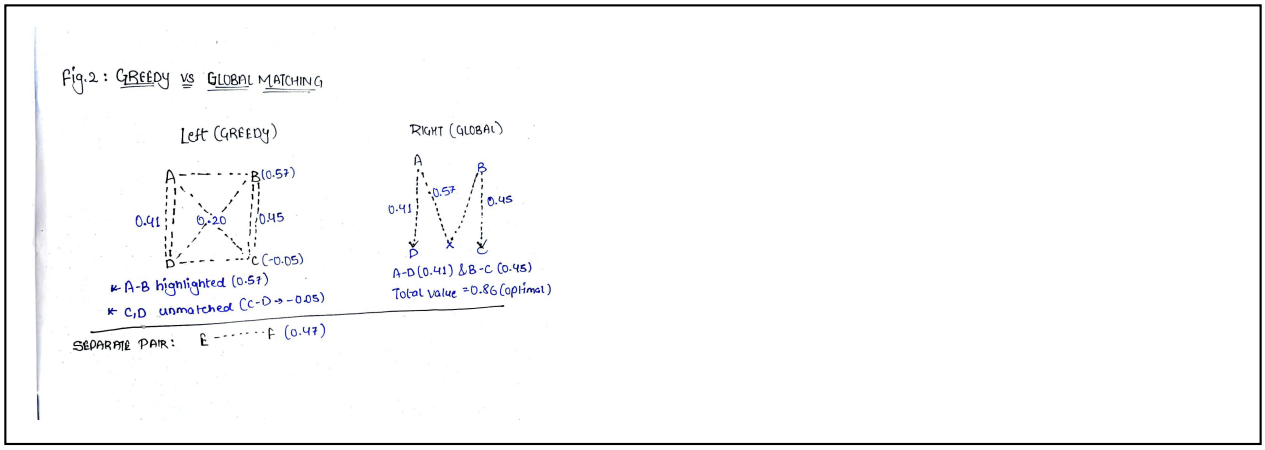


Figure 2. Greedy versus global selection on the same feasible edges (illustrative values from Section 7). *Space reserved for a hand-drawn diagram. Suggested content: members A, B, C, D as nodes; edge A–B alone (the strongest edge) versus A–D plus B–C, with adjusted weights written on the edges.*

Literature grounding and transfer limits. The method is informed by research, but none of the cited papers validates this policy or the synthetic outcome model. Edmonds' general-graph maximum-weight matching is a direct mathematical analogue for one day's weighted, non-overlapping batch; it solves a static subproblem, not the 60-day problem with arrivals and waiting [6]. Online matching research, including the Karp–Vazirani–Vazirani bipartite-arrival model and a later survey, studies arrivals under specific graph and information assumptions; those competitive guarantees do not automatically transfer to this shared-pool, reciprocal, daily-batch setting [7, 8]. Saar-Tsechansky and Provost motivate asking for fields that can alter a decision rather than only improving prediction accuracy; this motivates the value-of-information heuristic below but does not show that the clarification policy improves MSMI [9]. Lakkaraju et al. motivate caution because outcomes are observed only for assigned pairs [10]. Calibration work and the Brier score motivate held-out reliability checks for optional probability diagnostics, but do not remove selective-label bias [11, 12].

5. Clarification policy

The ask phase precedes matching. The daily budget is 12 units: a member's hard-constraint bundle costs 3 and a named soft field costs 1. The bundle is a simulator abstraction, not a real-service burden estimate. Clarification is immediate in version 1.0: a disclosed answer returns as a simulated self-report, while a declined answer stays unresolved. The policy never re-asks an observed or declined field and follows the status and ask-log contract [1–3].

Decision-focused value of information.

```
VOI(a) = ( E[ J(s after response to a) ] - J(s now) ) / cost(a)
```

J(s) is the total value of the best valid current matching; the expectation covers possible disclosed responses and refusal. Estimates must use training-rollout evidence, not hidden simulator values. For each eligible ask *a*, simulate plausible response states from declared, randomized clarification pilots within training rollouts, rebuild eligibility and the best matching under each possible response, and estimate the difference from the no-ask decision. A lower-cost question wins only when it yields greater decision value per unit, not merely greater predictive certainty. This is a proposed approximation, not a validated gain [9].

Budgeted ask selection.

1 Reject unaskable requests, already answered or declined fields, and requests for unavailable members.

2 Prioritize hard bundles that could unlock otherwise promising reciprocal edges, provided no known incompatibility already rules those edges out. Estimate gain by how much the refreshed answer could improve the best feasible pool matching, not by how many raw candidate pairs mention the member.

3 Consider soft questions only after hard feasibility is settled, for already feasible options, and only if the answer could change the edge ordering or the chosen batch. Divide expected improvement by the 1-unit cost.

4 Select a batch of requests within 12 units, respecting the interface's exact field names. Test joint bundles as well as single asks: two missing members can have near-zero individual VOI but positive value together.

Fallback if pilots are sparse. Deterministically rank unresolved hard bundles by the number and estimated quality of potentially unlockable alternatives, with overlap-aware discounting and deterministic tie-breaking, and use soft requests sparingly. This is reported explicitly as a heuristic, not an unbiased VOI estimate.

Example of not buying irrelevant information. Suppose member E could pair with F if E reveals acceptable zones. Asking E's constraints bundle (3 units) might open one feasible edge, whereas an unrelated soft question about someone with no eligible partners would not change today's matching. The policy may spend 3 of 12 units and leave the rest unused. If E declines, the E–F edge stays blocked [1, 2].

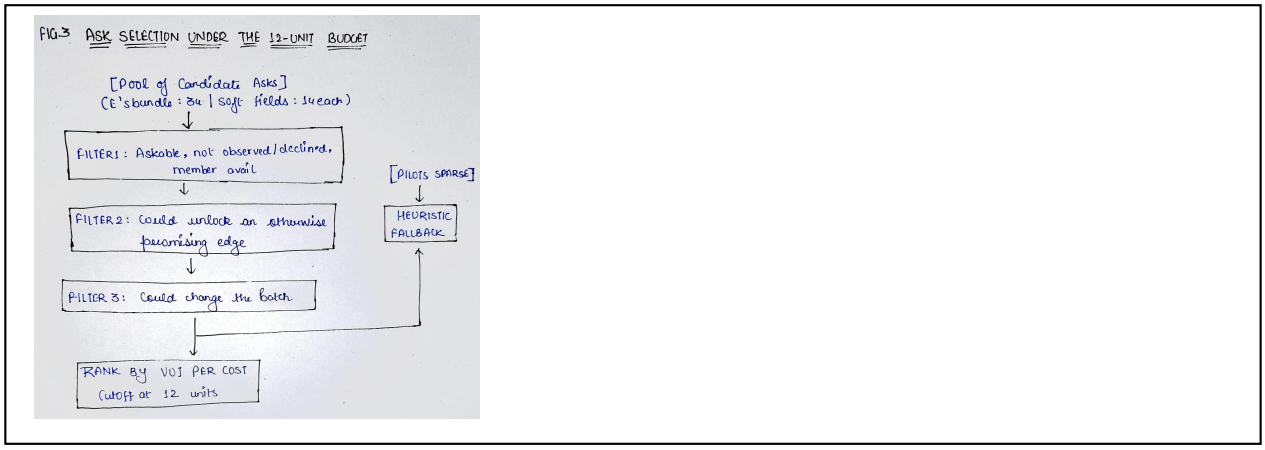


Figure 3. Ask-phase selection under the 12-unit budget. *Space reserved for a hand-drawn diagram. Suggested content: candidate asks, the three filters (askable, could unlock an edge, could change the batch), the VOI-per-cost ranking, and the budget cut-off.*

6. Allocation and abstention

Edge value.

For each fully feasible pair (i, j), compute q(i,j,t) using only observations available at decision time: shared relationship goals, pace, lifestyle, communication preferences and released feedback. A transparent initial score uses mutually observed soft features and smooths sparse evidence toward a neutral prior [1, 2, 10].

```
q = [k / (k + n)] * q0 + [n / (k + n)] * observed_similarity
```

Here n is the effective observed feature weight, k is a tuned shrinkage constant and q0 is a neutral baseline. Unknown fields are excluded from observed_similarity, not treated as mismatches. This is a ranking score, not a success probability. If mature, policy-generated rollouts provide enough data, a regularized, validated model of expected full MSMI is tested in its place.

Reservation value and graph optimization.

```
w(i,j,t) = q(i,j,t) - r(t)
```

r(t) is an opportunity-cost threshold tuned only on validation rollouts. Fixed, time-aware and observable pool-state-aware variants are compared; no future arrivals or outcomes are read. Build an undirected general

graph with one vertex per eligible, unoccupied member and one edge per fully feasible pair, weighted by w. Choose a deterministic maximum-weight matching, allowing vertices to remain unmatched, and drop non-positive edges [6–8].

```
maximize sum over pairs of w(i,j,t) * x(i,j)
subject to sum_j x(i,j) <= 1 for every member i, x(i,j) in {0, 1}
```

All candidate edges must pass the hard gate, and the empty matching is feasible. This optimizes one day's proxy objective; it does not claim globally optimal decisions across 60 days.

Abstention is an active action.

Return an empty pair list if no verified edge has positive adjusted value, or leave individual people unassigned if they are absent from the optimal disjoint batch. A higher early reservation threshold may preserve option value, and a lower threshold later may reduce unserved members. Both effects are hypotheses to test, not assumed truths [1, 7, 8].

Determinism, safety and runtime.

Use explicit tie-breaking so identical observable inputs and memory produce identical actions, validate the entire batch against the reference rules, and output valid empty lists on empty pools or empty graphs. Keep only compact, policy-owned JSON memory, reset between episodes; the policy is invoked afresh for each phase and does not assume process or filesystem persistence [2, 3]. A pinned general-graph blossom implementation is a suitable Round 2 candidate for the exact solver; its startup-inclusive runtime at the 200-member scale should be benchmarked before committing. If runtime or dependency limits make it unsafe, a deterministic greedy-plus-local-swap alternative is compared, and the policy never silently falls back to an invalid batch [1, 3, 5, 6].

Optional probability diagnostics.

Probabilities are reported only if support and calibration justify them, and they do not affect ranking. The four estimands are distinct: p_A_yes7 (A records "yes" to this assigned introduction by the seven-day response deadline), p_B_yes7 (the corresponding probability for B), p_mutual_yes7 (both record "yes" by the deadline, estimated directly where possible) and p_MSMI (the assigned pair satisfies the full MSMI definition). The product p_A_yes7 × p_B_yes7 is not used by default because it requires a conditional independence the specification does not establish. Calibration (reliability bins and Brier score) is reported on held-out, mature assignments and is scoped to assigned pairs under the tested policy, not all feasible pairs. Because historical logging propensities are null, no unbiased inverse-propensity evaluation is claimed from the snapshots [1, 2, 10–12].

7. Worked decision example

This miniature example shows the decisions the proposed policy would take. Every score and answer is hypothetical, not a simulator measurement or a learned compatibility probability.

Initial observation: six members. A, B, C and D have fully observed hard constraints. E and F would otherwise be operationally available, but E's acceptable-zone information has not been disclosed.

Pair	Current status	Soft ranking value q	Adjusted value w = q - 0.25
A–B	Feasible	0.82	0.57
A–D	Feasible	0.66	0.41
B–C	Feasible	0.70	0.45
C–D	Feasible	0.20	-0.05 (excluded)
E–F	Needs E's hard bundle	0.72 if feasible	0.47 once verified

- **Ask.** E's hard-constraint bundle has positive estimated decision value; request it for 3 units. No other ask is judged worthwhile today.
- **Refresh.** Assume E discloses a compatible zone preference; the eligibility engine now verifies E–F. If E declines or reveals incompatibility, E–F is excluded.
- **Score.** With an illustrative reservation value r = 0.25, the positive adjusted edges are A–B (0.57), A–D (0.41), B–C (0.45) and E–F (0.47); C–D (-0.05) is excluded.
- **Select.** The batch A–D + B–C + E–F has total adjusted value 0.41 + 0.45 + 0.47 = 1.33, and each of the six people appears once. The strongest single edge, A–B, is not chosen: it would block both A–D and B–C, and A–B + E–F totals only 1.04.
- **Alternative outcome.** If E's answer cannot verify feasibility, match A–D and B–C while E and F wait. If no positive adjusted edge exists, propose no introduction. This protects hard constraints and keeps invented answers out of decisions.

```
ASK(state): rank valid unresolved questions by matching-aware VOI; choose a joint
batch within the 12-unit budget
MATCH(new_state): build only fully feasible, eligible edges; score w = q - r; find
the maximum-weight non-overlapping positive edges; validate the batch; return pairs
(possibly an empty list)
```

8. Missingness, delayed data and leakage controls

Preserve the distinction among *observed*, *not_asked* and *declined*; a null is unknown, not zero or rejection. A missing feedback value before its deadline may be pending; after the response deadline, "no response" means no recorded response event for that target, not proof of dislike. A pair without a date has no observed second-meeting preference [1, 2].

Assignments, directional responses, mutual acceptance, date occurrence and each person's second-meeting intention are treated as separate events. Late "yes" answers stay in the record but do not count as timely MSMI. An immature assignment is not labelled a failure; it is evaluated after its observation window matures, and unresolved or missing feedback is reported separately [1, 2].

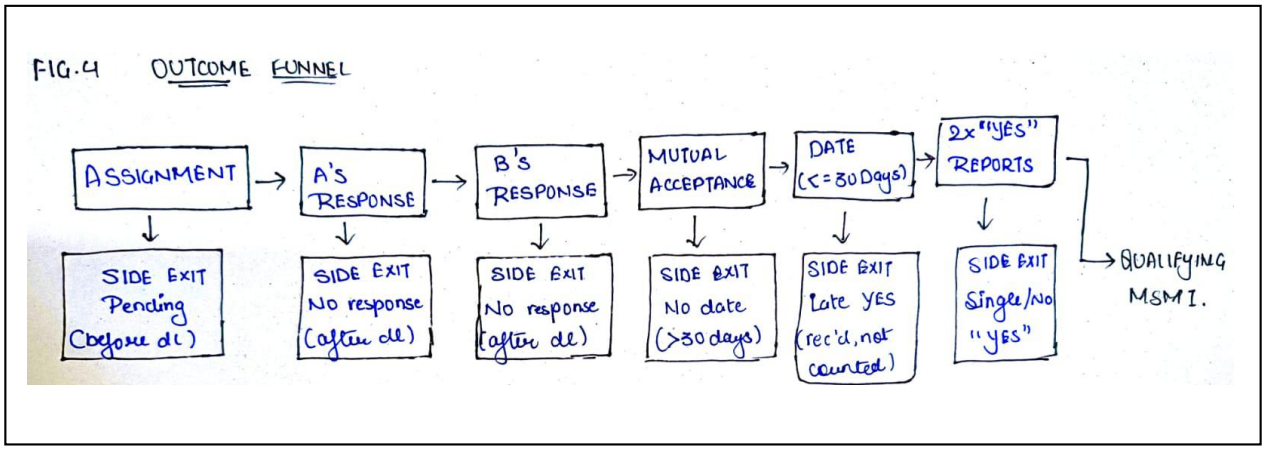


Figure 4. The outcome funnel from assignment to a qualifying MSMI event. *Space reserved for a hand-drawn diagram. Suggested content: assignment, directional responses (A and B), mutual acceptance, date within 30 days, two timely second-meeting "yes" reports within 3 days, with pending and no-response branches marked.*

Only fields whose observation day is no later than the action day are used, together with current refreshed clarification results, observable feedback, policy memory and declared training assets. The day-30 snapshot is not a valid substitute for earlier-day observations, and decision-time features are trained from day-zero rollouts, not from final profiles attached retrospectively [1–3].

Feedback is selectively observed on pairs the policy assigns [10]. Because historical propensity is null, offline comparisons of unassigned pairs are not identified as unbiased; complete policies are compared in controlled

simulator episodes and any extra assumptions are stated. Hidden simulator truth is never inspected, public seeds and identifiers are never used to reconstruct it, and future events are never used [1–3].

9. Evaluation plan

Splits and controlled comparisons.

The six designated training pools, two validation pools and two development-test pools are kept separate; rows from the same people or pairs are never split across them. Day-zero simulator rollouts are used for temporal training and policy comparison. Validation pools select model regularization, reserve values and clarification thresholds, and those choices are frozen before the two development-test pools are used for a final internal check. Static day-30 snapshots are for current-state exploration only [1, 4, 5].

The complete policy is compared with the supplied greedy, no-clarification and random-feasible baselines on identical seeds and public variants. The six public variants are `development`, `sparse`, `cold_start`, `delayed`, `shift` and `drift`; they are public stress tests, not access to the private assessment worlds [1, 4, 5]. At least 20 predeclared public seeds per variant are run where compute permits (120 episodes per method); otherwise the complete fixed seed list is disclosed and favourable runs are not selected. Methods are paired by seed and variant. Uncertainty is estimated by resampling matched episode-seed units within each variant and recomputing the equally weighted variant-mean difference; members are not resampled as if independent. An interval overlapping zero is treated as inconclusive, not as equivalence.

Planned ablations.

Comparison	Question answered
No clarification vs targeted VOI asks (also vs a hard-constraints-first heuristic)	Do purchased answers create enough decision improvement to justify their cost?
Greedy edge selection vs global maximum-weight matching, same edge scores and abstention value	Does pool-wide allocation outperform independent edge selection?
No reservation vs tuned reservation	When does waiting help rather than reduce introduction coverage?
Soft score vs learned outcome estimate; directional product vs direct mutual estimate (only if enough mature outcomes exist)	Is a trained MSMI estimate better once mature labels exist, and does the independence assumption hold?

Metrics and reporting.

The release's exact metric definitions are used. For each episode, N is the number of members arrived by decision day 59 (including people later paused or departed) and M is the number of qualifying introductions. MSMI per 100 = 100 × M / N, and coverage = distinct arrived members introduced / N; both are zero when N is zero. One qualifying pair contributes one MSMI outcome. Reported quantities: assignments, mutual acceptances, dates, qualifying MSMI count and MSMI per 100 arrived members, distinct-member coverage, clarification cost, missing and pending feedback, first-introduction waiting-time distribution and unserved count, inference time, all tested seeds, scenario-level results with paired uncertainty, validity failures (target zero) and cost–coverage trade-offs [1, 5].

The primary public comparison is the equally weighted mean of variant means, a transparent analogue of the official private score, which gives equal weight to six scenario-family means rather than pooling members; it is not a claim to reproduce the private assessment. Denominators are not pooled, no best seed is reported, and equivalence is not implied from small or zero MSMI counts. The official ranking tie-breaks are coverage, mutual acceptances per 100, lower ask cost, then lower inference time [1, 4, 5].

10. Expected failure cases and safeguards

Failure mode	Why it matters	Planned response / reporting
Sparse feasible graph	Some members have no eligible edge.	Leave unmatched; report coverage, unserved count and observable blockers.
Declined hard information	Missing hard facts block eligibility.	Never infer or bypass; report declines and blocked options.
Candidate competition	A top edge can block a better batch.	Use pool-wide matching; compare with greedy.
Selective, sparse outcomes	Labels exist only for assigned pairs; MSMI may be rare.	Evaluate full rollouts; calibrate only mature assigned cases; fall back to a non-probability score.
Distribution shift	Stress variants change supply, missingness, timing, weights or responses.	Report each variant; do not tune only on `development`.
Delayed or censored feedback	Recent outcomes are not yet observable.	Use released feedback only; separate pending, no-response and mature outcomes.
Myopic daily decisions	Today's matching cannot know future arrivals.	Test and ablate the reserve value; state that the policy is not horizon-optimal.
Over-waiting	A reserve may lower introduction coverage.	Compare against no reservation; report coverage and waiting time.
Runtime or protocol failure	One invalid episode makes the entry ineligible.	Test edge cases, batch validity, determinism and container runtime before freeze.

11. Scope and provenance

This note describes a proposed research plan. No policy implementation, experiment, benchmark result, calibrated model or superiority claim is asserted. It is scoped to the frozen synthetic release 1.0.0 and the observable JSON contract. Any later implementation will document dependencies, licenses, training seeds, the inference seed and deviations from this plan [1, 3, 4]. Pinned source commit checked 9 October 2026: `a8e26b35118cfa8e886a02f93984923e43ab64f6`. Later official corrections supersede this snapshot.

References

[1] Vouchsafe. *The One Introduction Problem*, participant specification 1.0.0 (5 October 2026). Problem statement.

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